

IP Addresses

What is an IP Address?

An IP address (short: IP, which stands for Internet Protocol) is the unique address of a device in a network. Similar to a postal address, it enables directed data exchange between network components. This ensures that data packages reliably reach their destination without confusion. Today there are IPv4 and IPv6 addresses. The v4 and v6 indicate the version number. There was experimentation with v5, but it was never released.

Structure of an IPv4 Address

An IPv4 address is a 32-bit long number, divided into four bytes (or four octets). Each octet can take values from 0 to 255 (**Binary: 00000000 - 11111111**). We therefore have 2^{32} or roughly 4.3 billion IPv4 addresses. IPv4 addresses always consist of a network portion and a host portion. Where the network portion ends and the host portion begins is determined by the subnet mask.

Classes of IPv4 Addresses

Class	Range	Network portion	Example use
A	1.0.0.0 - 126.255.255.255	First octet	Large enterprises / Providers
B	128.0.0.0 - 191.255.255.255	First two octets	Medium networks, e.g., Universities
C	192.0.0.0 - 223.255.255.255	First three octets	Small networks, e.g., LAN
D	224.0.0.0 - 239.255.255.255		Multicast (group communication)
E	240.0.0.0 - 255.255.255.255		Reserved (experiments)

Special IPv4 Addresses

0.0.0.0	Default route / "all addresses"
127.0.0.1	Loopback address (local machine)
255.255.255.255	Broadcast to all devices on the Internet

Fun Fact: 127.0.0.1 is not the only loopback address. It is actually **127.0.0.0/8** or the IP range **127.0.0.1 - 127.255.255.254**, providing over 16 million possibilities to "ping" oneself.

Private IPv4 Address Ranges

There are three private address ranges in IPv4, designated for private use only and cannot be routed on the internet.

10.0.0.0 - 10.255.255.255 Class A
172.16.0.0 - 172.31.255.255 Class B
192.168.0.0 - 192.168.255.255 Class C

IPv4 Address Exhaustion

IPv4 was introduced in the 1980s. With its 32-bit space and over 4 billion addresses, it was initially considered more than sufficient. So the inventors were very generous in assigning Class A networks with about 16.7 billion addresses each to large companies and institutions such as General Electric, IBM, HP, the MIT, as well as to the military and intelligence services.

By the 1990s, it became clear that the addresses were running out, and countermeasures began. One of the most important was the introduction of Classless Inter-Domain Routing (CIDR), which removed the rigid class system

and allowed for more flexible use of address space. Some companies even voluntarily returned their Class A networks to extend the lifetime of IPv4.

In addition, NAT (Network Address Translation) was introduced, along with the definition of the private IPv4 address ranges still commonly used today.

What does CIDR do?

Classless Inter-Domain Routing uses flexible prefix notation (e.g., /24) to define networks of any size, instead of fixed classes. This allows for precise address allocation and aggregation of large blocks, reducing the size of the global routing tables. The prefix indicates how many bits of the address, counted from left to right, are fixed. For example, in the network **192.168.0.0/24**, **192.168.0.** remains the same, and only the last number varies.

192.168.1.x would then be another network. For more details, see Subnetting.

What does NAT do?

In home and business networks, IP addresses from private IPv4 ranges are used. Access to the internet typically occurs via a router, which receives a public IPv4 address from the ISP (Internet Service Provider). The router uses Network Address Translation (NAT) to translate the private IPs of the local network into the router's public IP when sending data to the internet, and vice versa when receiving data from the internet. This allows multiple devices in the local network to access the internet simultaneously without requiring a public IP address for each device. They all share the router's IP and the router ensures the data reaches the correct recipients.

NAT only a temporary Solution?

The introduction of NAT was an important short-term measure to mitigate IPv4 address exhaustion.

However, as the number of internet-enabled devices merely exploded and NAT also has some disadvantages (e.g., higher complexity for applications with peer-to-peer communication), IPv6 was eventually introduced.

Structure of an IPv6 Addresses

An IPv6 address is a 128-bit long number, divided into eight blocks of 16 bits each, separated by colons. A block is often called hextet but may also be referred to as a segment, quartet, or 16-bit block, though there is no universally accepted naming. Each hextet has four hexadecimal digits and can take values from **0000** to **ffff** (**0** - **65535**), with each digit representing a 4-bit binary value **0000** - **1111** (**0** - **15**). A typical IPv6 address looks like this:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

With this version of the internet protocol we now have 2^{128} IPv6 addresses, which is:

340,282,366,920,938,463,463,374,607,431,768,211,456. This is more than 10^{28} IPs per human on Earth (with 8 billion people currently).

IPv6 Address Shortening

Leading zeros in a hextet can be omitted. The above IPv6 becomes:

2001:db8:85a3:0000:0000:8a2e:370:7334

Additionally, blocks consisting entirely of zeros may be omitted (but only once per address). They are replaced by two consecutive colons ("::"), which represents the omitted zero block. For example:

2001:db8:85a3::8a2e:370:7334

If the zero block is at the beginning or end of the address, the double colon remains at the edge:

- **fe80::** (omitted zeros at the end)
- **::1** (omitted zeros at the beginning)

IPv6 Address Composition

An IPv6 address typically consists of a 64-bit network prefix (similar to the network portion in IPv4) followed by a 64-bit interface identifier (unique identifier for the host in the network, except Anycast). The interface identifier is often generated from the MAC address or is completely random.

Types of IPv6 Addresses

There are different kinds of IPv6 addresses, some of which can be identified by their prefix.

One type is Unicast addresses, which are further divided into Global Unicast, Link-Local, and Unique Local.

Additionally, there are Multicast and Anycast addresses.

Unicast Addresses

Global Unicast	Publicly routable on the internet. Range: 2000:: - 3fff:ffff:ffff:ffff:... Short: 2000::/3 Currently mostly 2000::/4
Link-Local	Valid <u>only</u> in the local network. Range: fe80:: - febf:ffff:ffff:ffff:... Short: fe80::/10 Typically fe80::/64 is used.
Unique Local	Similar to private IPv4 addresses. Range: fc00:: - fdff:ffff:ffff:ffff:... Short: fc00::/7 fd00::/8 is set as standard.

Multicast

Multicast is a core concept in IPv6, replacing IPv4 broadcast addresses with efficient group communication. Unlike broadcasts, which burden all hosts in the network, or unicast (one-to-one communication), multicast allows targeted addressing of recipient groups (one-to-many or many-to-many) that subscribe to specific services or messages.

Multicast Address Format

The theoretical address range for multicast is ff00::/8. However, the third "hex" (i.e., the third digit or the bits 9 - 12) acts as a flag (like a "signal flag") that provides additional information about the multicast address. The fourth "hex" (the fourth digit or the bits 13 - 16) represents the scope, which defines the address's validity domain.

8-bit	4-bit	4-bit	112-bit
ff	Flag	Scope	Group ID

Flags 4-bit

Value	Meaning	Description
0	Permanent (well-known)	Defined by IETF (see note below)
1	Transient (temporary)	Dynamically created, e.g., by applications
2	Not used	Reserved
4	Rendezvous Point (RPL)	For routing protocols

The flags 3 and 5 - f are undocumented, meaning they are either free for use or reserved for future purposes.

Note: The Internet Engineering Task Force (IETF) is a major organization focused on the technical evolution of the Internet, with the goal of developing Internet standards and so-called best practices.

It comprises network engineers, manufacturers, network operators, researchers, and users - participation is open to anyone. Through regular meetings, the IETF forms working groups dedicated to specific topics like protocols, security, or network technologies.

The outcomes of the IETF's work are published in documents known as RFCs (Requests for Comments). These include both purely informational RFCs and those with binding authority, which are recognized as official standards (Standards-Track RFCs). Although these standards are not legally mandated, failure to adhere to them risks incompatibility with established systems.

For example, IPv4 was formally defined in the binding RFC 791 (1981), while IPv6 was introduced via RFC 2460 (1998).

Scopes 4-bit

Hex	Name	Description
1	Interface-Local	On the device itself
2	Link-Local	Within the local network segment
3	Subnet-Local	Within the same subnet (deprecated)
4	Admin-Local	Within an administrative domain
5	Site-Local	Within an entire site (e.g., campus)
8	Organization-Local	Within an organization
E	Global	Internet-wide

The scopes 0, 6, 7, 9-D, and F are unused and/or reserved for future purposes.

Note: `ff02::1` reaches all nodes in the local network, where a node is any device using IP addresses (printers, servers, IP phones, IoT devices), while an interface refers specifically to the node's network interface.

Anycast

Anycast assigns a single IPv6 address to multiple devices (typically servers or routers). These devices are located in different parts of the network or distributed worldwide.

When a device sends a packet to such an anycast address, it will always be routed to the "nearest" node according to the routing protocol. The routing protocol determines what qualifies as "nearest", based on criteria such as hop count, latency, or administrative costs.

Important: The address looks like a normal unicast address. Anycast is determined by routing behavior, not by a special address range or prefix.

Advantages of Anycast

- High availability: If one node fails, another automatically takes over.
- Load balancing: Requests are distributed across servers transparently.
- Low latency: Clients reach the "nearest" server automatically.
- Simple addressing: Services like DNS remain reachable under the same IP globally.

Limitations of Anycast

Anycast is well-suited for stateless protocols like UDP (e.g., DNS)

With TCP connections, issues may arise if the connection is redirected to a different anycast node during the session, which for example can occur during connection drops or routing changes. This can also cause instability with protocols like HTTPS, SSH, or FTP.

Examples of Anycast in Practice

Public DNS Services

The DNS servers from Google have the IP: `2001:4860:4860::8888`

The ones from Cloudflare have the IP: `2606:4700:4700::1111`

These IPv6 addresses are globally reachable, but the nearest node always responds.

Content Delivery Networks (CDNs)

Providers like Akamai, Cloudflare, and Fastly use anycast to deliver web content fast and redundantly.